# Module 3
# Business Economics

# Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

## SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

### Answer 1(a)

A substitute is a good or service whose increasing consumption would replace the consumption of another (related) good or service.

A complement is a good or service whose increasing consumption results in the consumption of more of another (related) good or service.

### Answer 1(b)

(i) Factors leading to changes in demand (shift in demand curve):
- Change in income
- Change in taste
- Change in the price of related goods (substitutes and complements)
- Change in population

[Any three of the above]

(ii) Factors leading to changes in supply (shift in supply curve):
- Change in input prices
- Change in natural condition (e.g. persistent drought that affects harvest)
- Change in (new) technology
- Change in government regulation, tax or subsidies

[Any three of the above]

[Other valid answers are acceptable.]

### Answer 1(c)

The assumption behind a demand curve is that all relevant economic factors, other than the product's own price, are held fixed. It follows that changes in any of the non-price factors would affect the quantity demanded at any given price level, resulting in a shift of the curve.

### Answer 1(d)

The opening of new theatres would increase the supply (quantity supplied) of theatre seats (at any given price level) and thus shift the supply curve of movie theatre seats to the right.

### Answer 1(e)

Snacks and movie-watching are complements (in consumption). A significant rise in the price of snacks would decrease the quantity demanded of snacks and thus the demand for movie tickets, resulting in a leftward shift in the demand curve for movie tickets.

**Answer 1(f)**

Cross-price elasticity for complements (movie tickets and snacks) is negative in value.
Cross-price elasticity of demand = % change in quantity demanded for movie tickets / % change in price of snacks
% change in quantity demanded for movie tickets = -0.7 x 30% = -21%
The movie tickets sales will go down by 21%.

**Answer 1(g)**

| | **Direction of Price Change** | **Direction of Quantity Change** |
|---|---|---|
| **Event (I):** | P* goes down | Q* goes up |
| **Event (II):** | P* goes down | Q* goes down |
| **Net effect:** | P* goes down | Q* could go up or down depending on the relative strength of the demand and supply curves' shifts |

**Answer 2(a)**

Inflation refers to an increase in the general prices of goods and services.

Stagnation refers to an extended period of slow, no or even negative growth in an economy (as measured by gross domestic product) with high unemployment rate.

Stagflation refers to the situation in which the economy is experiencing both inflation and stagnation at the same time.

**Answer 2(b)(i)**

Overall, CPI is considered as a more appropriate measure of the consumers' cost of living as it includes (domestically produced and imported) goods and services consumed by a typical household.

**Answer 2b(ii)**

The CPI tends to overstate inflation and thus cost of living as it ignores the fact that consumers tend to substitute lower-priced alternatives for higher-priced goods/ services. Moreover, technological advances over time tend to increase the life and usefulness of products.
[Other valid answers are acceptable.]

**Answer 2c(i)**

Distribution effect: Unexpected inflation redistributes wealth and income from one group of economic agents to another, meaning that some groups may gain while others may lose.

Deadweight loss effects:

- Loss of purchasing power – Inflation degrades the value of currency, which makes it harder for consumers to purchase the same amount of needed goods and services over time (hence lower standards of living).

- Lower real wage growth – Inflation erodes real wage growth rates for individuals, causing people to spend less.

- Slower GDP growth – Because inflation makes goods and services more expensive and causes lower real wages, people will consume less, which slows the rate of GDP growth.

- Higher unemployment – As inflation slows the rate of GDP growth, unemployment rates increase. Firms will adjust output in response to declining demand.

- Decreased international competitiveness – Inflation increases the prices of goods and services, making a nation's exports less competitive in the international marketplace.

- Economic uncertainty – Inflation causes uncertainty about future prices, costs and profit margins for businesses. This uncertainty makes businesses less likely to invest capital, slowing GDP growth.

[Any ONE of the above]
[Other valid answers are acceptable.]

**Answer 2(c)(ii)**

- Borrowers (debtors) who are paying a fixed rate of interest will gain because the real value or purchasing power of the fixed interest paid falls. Lenders (creditors) receiving a fixed rate of interest will lose (or more generally, interest rates are hard to keep up with inflation rates) because the real value or purchasing power of the fixed interest received drops.

- Savers (depositors) will lose because the real value or purchasing power of the fixed interest received drops. Bankers will gain because the real value or purchasing power of the fixed interest paid drops.

- Life insurance policyholders will lose because the real value or purchasing power of the fixed amount of compensation received drops. Insurance companies will gain because the real value or purchasing power of the fixed amount of compensation paid drops.

**Answer 2(d)(i)**

A supply chain disruption refers to a breakdown in the manufacturing flow of goods and their delivery to customers. It will cause a leftward shift of the AS curve, resulting in lower output and higher price.

**Answer 2(d)(ii)**

Demand-pull inflation occurs when aggregate demand is increasing at a faster pace than the aggregate supply.

Cost-push inflation is driven by a decrease in aggregate supply and is usually due to external factors such as natural disasters and oil supply shocks.

**Answer 2(d)(iii)**

Stagflation is essentially stagnant/slow economic growth plus high inflation and high unemployment.

As monetary policy affects the economy through influencing the aggregate demand, it can normally help reduce inflation (by raising interest rates) or stimulate economic growth (by cutting interest rates). It follows that monetary policy cannot solve both inflation and recession at the same time.

**Answer 3(a)**

Sampling is the selection of individual observations to form a subset of the population to estimate characteristics of the population.

Since it is impossible to observe the characteristics of the population in most cases, sampling allows researchers to draw inference about the population based on the sample data.

**Answer 3(b)**

Raw data => **(random) sampling** => sampled data => statistical inference => **population (parameter)** => Information

**Answer 3(c)**

Time series data consists of data points of a single subject recorded at regular time intervals over a given period.

Cross sectional data consists of data points of many subjects at the same point in time.

**Answer 3(d)**

The four components of time series are:

1) Secular trend describes the long-term pattern of a time series.
2) Seasonal variations describe a repetitive cyclical pattern within a year.
3) Cyclical fluctuations describes the repetitive up-and-down movement around a trend that persists over multiple time periods.
4) Irregular variations refer to the random shocks (or changes) that follow no apparent pattern.

**Answer 3(e)**

Q3/2021 (yoy) growth rate = ($5b-$3b)/$3b = 0.67 (67%)

Q4/2021 (yoy) growth rate = ($6b-$2b)/$2b = 2.00 (200%)

Q3/2021 (qoq) growth rate = ($5b-$4b)/$4b = 0.25 (25%)

Q4/2021 (qoq) growth rate = ($6b-$5b)/$5b = 0.20 (20%)

**Answer 4(a)**

Sampling error is the difference between a sample statistic used to estimate a population parameter and the actual (but unknown) value of the parameter.

A confidence interval gives an estimated range of values (calculated from a given set of sample data) that is likely to include an unknown population parameter (e.g. mean).

**Answer 4(b)**

The Statement is false.

An estimate of the population parameter such as population mean based on a random sample is unbiased. With the size of sample increases, the estimate would be more precise.

**Answer 4(c)(i)**

p = 480/500 = 0.96, n=500, $Z_{0.05}$ = 1.96

$$= \left(0.96 - 1.96\sqrt{\frac{0.96(1-0.96)}{500}}, 0.96 + 1.96\sqrt{\frac{0.96(1-0.96)}{500}}\right)$$

= (0.96 – 0.0172, 0.96 + 0.0172)
= (0.942, 0.9772)

**Answer 4(c)(ii)**

False. Just as with any statistic estimated from a (random) sample, the upper and lower bounds of the confidence interval will vary from sample to sample. It follows that a 95% level of confidence means that 95% of the intervals calculated from the random samples will contain the true population mean.

* * * END OF EXAMINATION PAPER * * *